Text Statistics Tool

Contents

Statistical tools

The NLP Suite computes statistics through two dedicated GUIs — Statistics (text) for raw corpora and Statistics (csv) for tabular data (including the csv tables produced by other NLP Suite tools) — plus association statistics for n-grams and collocations. All options produce csv output and charts.

Statistics on text corpora — "Compute corpus statistics"

- Frequencies of sentences, words, syllables, and top-20 words — basic corpus counts and the 20 most frequent words.
- Sentence length — distribution of words per sentence.
- Line length — distribution of words per line (useful for verse, transcripts, structured text).
- TF-IDF (most distinctive words per document) — term frequency–inverse document frequency: surfaces the words that most characterize each document relative to the rest of the corpus (common words are down-weighted).
- ***Lexical diversity*** — how varied the vocabulary is:
  - TTR (Type-Token Ratio) — types ÷ tokens; simple but sensitive to text length.
  - MTLD (Measure of Textual Lexical Diversity) — length-robust alternative to TTR.
  - vocd-D — diversity estimated by curve-fitting; also length-robust.
- ***Readability scores*** — how hard the text is to read:
  - Flesch Reading Ease — higher = easier.
  - Gunning Fog — approximate years of schooling needed.
  - Coleman-Liau — grade level from characters (not syllables).
- ***Word frequency distribution (Zipf's Law)*** — tests whether word frequencies follow the expected rank–frequency (Zipfian) pattern.

Text options: *Lemmatize words* (count lemmas rather than surface forms); *Exclude stopwords & punctuation*.

Also: *Compute corpus statistics by POS (Part-of-Speech) tag value* — the above, broken down per POS category.

Statistics on csv / tabular data

- ***Descriptive statistics on numeric field(s)***: Mean, Median, Mode, Sum, Minimum, Maximum, Count, Standard deviation, Variance — with an optional Group-by field (e.g. mean value per country/year).
- ***Frequencies of csv field(s)***: value counts for one or more columns.
- ***Hypothesis tests*** (non-parametric — they do not assume a normal distribution):
  - Group comparison — pick a numeric *value* column and a *group* column. The Suite chooses automatically:
    - Mann-Whitney U test for two groups (effect size: Cliff's delta).
    - Kruskal-Wallis H test for three or more groups (effect size: epsilon-squared), with Dunn's post-hoc pairwise comparisons (Bonferroni-corrected) to show *which* groups differ.
  - ***Log-likelihood (corpus comparison / keyness) — Dunning's G-test***: identifies words statistically over- or under-represented in one corpus versus another. Requires a *word* column plus either two *frequency* columns or a *corpus-ID* column.
  - **Chi-square test of independence** – tests whether two categorical columns are associated (pick two category columns); reports the Chi-square statistic, p-value, Cramér's V effect size and standardized residuals.

- Correction
    - **Correlation (Spearman / Kendall)** – tests monotonic association between two numeric columns; reports the coefficient, p-value, confidence interval and a fitted regression line.
    - **Mann-Kendall trend test** – tests for a significant increasing or decreasing trend in a numeric series over time (a date column); reports Kendall's tau and Sen's slope.
    - **Change-point detection (Pettitt's test)** – finds a single abrupt shift in a time-ordered numeric series (e.g. topic prevalence or sentiment) and reports where it occurs and the mean before and after the change.
    - **Permutation test (two groups)** – a distribution-free test of the difference in means between two groups, obtained by repeatedly shuffling the group labels; reports the observed difference, p-value and Cohen's d effect size.
    - **Inter-annotator agreement (Cohen's / Fleiss' kappa)** – measures how well two or more annotators or tools agree on labels (e.g. Stanza vs spaCy POS or NER tags); Cohen's kappa for two annotators (with a confusion matrix), Fleiss' kappa for three or more, with a Landis-Koch interpretation of the score.

Collocation / co-occurrence association statistics (N-grams & Co-occurrences)

For each candidate word pair, the strength of association is measured by:

- ***PMI (Pointwise Mutual Information)*** — how much more often the pair co-occurs than chance; favors rarer, tightly-bound pairs.
- ***Log-Likelihood*** — significance of the association; robust for low frequencies.
- ***Chi-Squared*** — significance via observed vs expected co-occurrence.
- ***T-Score*** — confidence in the association; favors more frequent pairs.
- ***Dice Coefficient*** — overlap-based association, independent of corpus size.

Input

- You can give in input a set of specific N-Grams to look for and it will compute the frequency of the input word lists. For example, the routine can search in all the text files in a folder to show how many times the words: "the court", "white men", "sheriff" appear in the files. Regular expressions (not wildcards) are supported (for more information see: https://docs.oracle.com/javase/7/docs/api/java/util/regex/Pattern.html).
- You can choose to multiply the frequency of each of the search terms by enclosing the term in "(" ")" and writing the scaling factor preceded by *, for example "(sheriff)*10".
- You can choose the following grouping options:
    - By date (same day, month, quarter or year)
    - All together: the statistics are computed as if all the articles were one
    - No grouping: each file has its own statistics
- The analysis can be case-sensitive

Output

- A csv file that contains the statistics mentioned above regarding the text files you have chosen to analyze.
- If you choose to group the articles by date (day, month, quarter, year) then the results will be organized in groups. In each group there will be all the text files with the chosen characteristic (same day, same month, …). All types of analysis will be performed on each group and all the results for each group will be printed in the same output file.

Output file overview

The output file is organized in different sections:

- Groups composition (group label - group components)
  In this section there are two columns, one containing the labels of each group, the other containing the list of the filenames that belong to that group.
  The group labels can be a number (if you choose to group the text files by year for example, the label will be the year), or a string (if you choose not to group the files then the group label will be the filename).
- Searched N-Grams
  This section contains in one comma separated list the N-Grams that have been searched in each group
- Searched N-Grams statistics
  This section contains the statistics regarding each searched N-Gram. The results are organized in a table where the firs column identifies the group for which the results apply, the other columns contains the frequency of each of the searched N-Grams.
- Words count
  This section contains the count of the words considered in each group, specifying also how many distinct words were found.
- Word length distribution
  This section contains the frequency distribution of the length of the words for each group. There are two columns, one for the word length, the other for its frequency.
- General N-Grams statistics, organized in:
  - 2-Grams frequencies
  - 3-Grams frequencies
  - 4-Grams frequencies
  - 5-Grams frequencies

  The results for each N-Gram are organized in two columns, one contains the N-Gram, the other its frequency. N-Grams with a frequency lower than 2 are not displayed.
- Word Frequencies
  This section contains the frequencies of all the words in each group. The results are organized in two columns, the firs one containing the word, the second one its frequency.

Each one of these sections has as many subsections as the groups are. In each of those subsections there are the global statistics regarding each group.

Pre-requisites for grouping by date

If your file name contains a date and it can be split into multiple parts, you can choose a separator and the position of the date part in the file name in order to aggregate the statistics of each text file by date.
For example, the file name "The Atlanta Constitution_01-12-1899_1.txt" could be a valid file name to be processed. In this case you would have to choose the separator that you used (in this case "_") and the position of the date object in the file name (in this case 1 because there is the newspaper name in the position 0, "The Atlanta Constitution").
The tool can group files by day, month, quarter (quarters are numbered from 0 to 3) or year.

You can also choose to get global statistics based on all files grouped together or to get separate statistics regarding each file.

Notes

- The tool removes the punctuation signs in each text file and considers all the words separated by one or more whitespace characters (space, tab, new line characters for example). If you choose to perform the analysis in a non-case-sensitive way, then all the text is converted to lowercase and then processed.
- Only plain text files are supported.
- All languages that separate words with one or more whitespace characters can run the analysis offered by this tool.